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May 28, 2025

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Reference Letter for Zheyao Zhu's NIW Petition

To Whom It May Concern,

I am writing to convey my strongest support of Zheyao Zhu's National Interest Waiver (NIW) petition. I had the privilege of working closely with Zheyao during his undergraduate studies where he contributed immensely to the decommissioning of facilities within the nation's Nuclear Weapons Complex. I will describe his uncanny technical abilities that were evidenced so early with me, and some of what I know since by following his work over time. I want to emphasize that the qualifications, credentials and experiences that I describe here meet an exceedingly high bar.

I am broadly known as the father of field robotics. The discipline pertains to thinking machines at work in the world. I've pioneered major contributions to robotics and automation as they relate to nuclear, farming, mining, construction, logistics, automotive, defense and space over a 46 year career. I innovated the world's first autonomous outdoor driving, then developed that with others to high functionality and infusion into government and industrial applications. I developed the robots and operations that cleaned up the nuclear accident at Three Mile Island. I innovated the precise in-pipe robotic assay which is fundamental to cleanup of the facilities where our nation enriched the U-235 uranium of the cold war era. This is the context under which I first met and worked with Zheyao. I won DARPA's 2007 Urban Challenge which ignited the automotive automation industry. I am the SCS Founders Research Professor at Carnegie Mellon University's #1-ranked Robotics Institute. There I founded the Field Robotics Center (FRC) and the National Robotics Engineering Center. These entities have been instrumental in advancing technologies and enterprises for all manner of applications contributing significantly to both academic research and practical applications in the field of robotics. I've spun out multiple companies. I am a member of the National Academy of Engineering. Distinctions include the Simon Ramo Medal for robotic systems, Columbia Medal for contribution to aerospace, and the Feigenbaum Prize for artificial intelligence. I am recipient of the Engelberger Award for robotics, the Metcalf award for engineering excellence, and the Newell and Teare Awards among others.

In my decades of experience in robotics education and research, I have mentored hundreds of engineers who have made significant contributions to the field of robotics and to the national interest. One notable example is Chris Urmson who played a pivotal role in my DARPA Urban Challenge racing team. Urmson later became a key figure in Google's self-driving car project and is currently co-founder and CEO of multi-billion dollar Aurora AI. John Bares is one of the most diverse robot developers of his time, led the National Robotics Engineering Consortium, led Uber's automation, and is currently founder-CEO of Carnegie Robotics. It is with this background that I can attest to Zheyao Zhu's technical abilities and potential for high-impact contributions in robotics and autonomous systems.

Zheyao Zhu's Contributions to the RadPiper Project

During his undergraduate research at CMU's Field Robotics Center, Zheyao made substantial contributions to the RadPiper robot. This is an autonomous robotic system for measuring uranium-235 deposits within the piping of America's Nuclear Weapons Complex. The U.S. Department of Energy (DOE) funded the project to aid in the safe decommissioning of former nuclear processing plants, reducing radiation exposures that formerly related to human measuring techniques. Secondary benefits included streamlining the operations and cost savings. Recognitions from the application facility, publications, conferences and coverage from media outlets garnered widespread recognition. These reports highlighted RadPiper's potential to broadly accelerate nuclear site cleanup efforts, replacing outdated, labor-intensive methods with a more efficient and autonomous solution.

Zheyao's specific contributions to the project were pivotal. He addressed the enabling and critical need for this robot system to precisely know its location within pipes. Additionally he developed means to extricate the robot from pipes in the eventuality of failure. The first is essential to any pipe inspection robot, and the latter is essential to deployments in high-consequence environments where failure is not an option.

(1) RadPiper was required to self-determine its precise location within piping so that its measurements of uranium could be correlated to position. The traditional robotic approach of visual odometry failed in the featureless pipes and no external positioning cues were available**.** What could possibly succeed? Zheyao conceived and developed an incredibly innovative Acoustic Odometry Localization System. My seasoned technical expertise was dubious that the approach could possibly succeed. Zheyao, however, persevered to exceed all expectations resulting in a low-cost, highly precise and reliable localization solution. I will remember forever the creativity, skill and work ethic that Zheyao brought to bear in his development of this important technology.

(2) RadPiper was tetherless. Anything that becomes stuck or lost within nuclear piping becomes part of the problem, not a solution. Zheyao and a fellow researcher conceived and developed a tethered robotic rescue system capable of extricating RadPiper should the need ever arise. This innovation significantly improved the operational resilience of the inspection process and ensured fail-safe deployment in high-risk environments.

RadPiper's Impact and National Importance

The RadPiper project represented a major advancement in autonomous infrastructure inspection and was a prime example of how robotics can contribute to national security, environmental protection, and industrial automation. Zheyao's contributions to the project not only improved robotic localization and reliability, but also supported a national priority-nuclear decommissioning and site safety. His research aligns with the U.S. Department of Energy's broader efforts to enhance autonomous systems for radiation detection and hazardous material management.

Beyond nuclear decommissioning, Zheyao's expertise in localization, robotic automation, and system redundancy has wide-ranging applications, including:

- Pipeline and Infrastructure Inspection - Similar robotic systems could be deployed for gas and oil pipeline inspections, reducing the risk of environmental disasters.
- Search and Rescue Robotics - His robotic recovery work is applicable to disaster recovery operations, where robots must operate in confined and hazardous environments.
- Autonomous Manufacturing Systems - Acoustic odometry techniques could improve localization in factories, warehouses, and autonomous logistics systems, advancing Industry 4.0 initiatives.

Zheyao Zhu's Post-Graduation Work and National Interest

Since completing his graduate studies, Zheyao has continued to make exceptional contributions to the field of robotics and automation. His post-graduation work directly advances the United States' leadership in automation, self-driving technology, and advanced manufacturing, fields critical to national security, economic competitiveness, and industrial automation.

At Embark Trucks, Zheyao was a key engineer in developing a next-generation factor graph-based six-degree-of-freedom (6DOF) odometry system that improved vehicle localization accuracy. His work on perception algorithms—including lidar-based lane line detection, barrier tracking, and radar calibration—enhanced the robustness and safety of autonomous trucking, a sector poised to revolutionize U.S. logistics and supply chain efficiency. His contributions continue to impact Applied Intuition, which acquired Embark's technology and now integrates it into U.S. military and off-road autonomous vehicle applications.

Following this, at Atomic Machines, Zheyao spearheaded innovations in MEMS (Micro-Electro-Mechanical Systems) fabrication automation, an area crucial for restoring U.S. competitiveness in advanced manufacturing. He developed computer vision-based pick-and-place robotics that enabled fully automated production cycles, increasing throughput by tenfold and significantly reducing human intervention. He also designed a cloud-based computer vision data pipeline to ensure scalability and optimization in high-tech manufacturing. Given the national priority to onshore advanced manufacturing, Zheyao’s work at Atomic Machines directly supported the United States’ efforts to regain dominance in production and industrial automation.

The National Importance of Pullover Reasoning in Autonomy

Currently, at Waymo, Zheyao is advancing the critical domain of pullover reasoning. This capability is far more than a convenience feature; it represents one of the most safety-critical and service-defining functions in autonomous driving. Pullover decisions affect passengers' first and last impressions of the service, determine accessibility for vulnerable populations, and ensure safety in both routine and emergency contexts.

From a safety standpoint, improper pullover locations—such as in the middle of busy streets—can put passengers at risk. In emergencies, such as hardware failures on highways, pullover reasoning must identify safe spots, sometimes requiring complex lane changes under degraded conditions. For elderly or disabled riders, convenient pullover locations are not just a matter of preference but a requirement for mobility. And from a traffic standpoint, poorly chosen pullover maneuvers can block lanes, create congestion, or impede emergency vehicles in dense urban environments.

The challenge is immense. Pullover reasoning requires balancing competing priorities: passenger convenience, traffic safety, compliance with laws, and overall flow of vehicles. It involves navigating ambiguous and dynamic real-world contexts such as airports and narrow city streets, where even human drivers must make nuanced judgments. Historically, pullover systems have been handcrafted and heuristic-driven. Transitioning to data-driven, learning-based systems marks a significant leap in the capability and reliability of autonomous vehicles.

Zheyao Zhu's Contributions at Waymo and Their Impact

Within this context, Zheyao's current role at Waymo is to help lead the transition from heuristic-based pullover logic to machine learning and reinforcement learning-based systems. This next-generation approach promises to generate more diverse and contextually appropriate pullover candidates, reason more effectively about tradeoffs, and produce more natural, human-like behaviors.

His contributions include designing data labeling policies for highly subjective "good" pullover locations, developing automated tools to generate meaningful datasets at scale, and building evaluation frameworks that define rigorous, safety-critical performance metrics. He is also working on reinforcement learning methods that adapt and improve with increasing data volume, enabling the pullover system to scale to new environments over time.

The expected impact is substantial:

- Enhanced passenger safety and convenience, directly advancing mobility for all demographics.
- Improved traffic flow in cities and airports by reducing blockages and congestion from stationary vehicles.
- Expansion of service areas into complex, high-demand environments such as airports.

- Increased reliability of emergency pullover maneuvers, protecting passengers during high-speed highway incidents.

Finally, Waymo is known for its public autonomous driving dataset, which has become one of the most widely used research resources worldwide. Zheyao’s work in advancing pullover reasoning will shape the data, metrics, and behaviors reflected in these releases, thereby influencing not only Waymo’s fleet but also the broader global research community. In this way, his work extends far beyond one company, reinforcing U.S. leadership in both the deployment and scientific study of autonomous systems.

Zheyao's expertise in robotic perception, computer vision, and real-time automation extends beyond his current roles, with applications in autonomous logistics, national defense, and smart manufacturing. His work aligns with the U.S. government's strategic initiatives in industrial automation, self-driving vehicle technology, and advanced manufacturing, all of which are critical for economic growth and national security. Given his strong technical expertise, research-driven approach, and the broad applicability of his work, Zheyao represents the type of exceptional engineering talent that directly supports U.S. leadership in robotics and automation. His contributions not only advance state-of-the-art technology but also help the United States maintain its competitive edge in the global technology race.

Conclusion

Throughout my career, I have worked with numerous students who have gone on to become leading figures in robotics and automation, and Zheyao Zhu stands out as one of the most promising researchers I have mentored. His work in robotic perception, planning, localization, and reliability engineering is directly aligned with national interests in nuclear safety, industrial automation, and autonomous systems. His ability to develop practical, field-deployable solutions to some of the most complex challenges in robotics demonstrates his value to the United States' technological advancement and innovation ecosystem.

Sincerely,

William Whittaker

William “Red” Whittaker
SCS Founders University Research Professor
Director, Field Robotics Center
Carnegie Mellon University